NUMS 2025

National University of Medical Sciences | Entrance Test | With Official Answer Key

National University of Medical Sciences (NUMS) Entrance Test 2025
150 Multiple Choice Questions
Physics (40) · Chemistry (40) · English (15) · Biology (55)

Time Allowed: 150 minutes | **Marking:** +1 per correct answer, no negative marking

Correct answer highlighted in **blue** under each question.

Recreated in a clean exam-style layout for self-practice. Content matches the original NUMS 2025 paper and its official answer key.

Section 1: Physics

Q1

The path of a charged particle in a plane perpendicular to a uniform magnetic field is:

- **A. Circular ✓ Correct**
- **B.** Helical
- **C.** Parabola
- **D.** Elliptical

Q2

Magnetic force acting on a charge at rest is:

- **A. Zero ✓ Correct**
- **B.** Positive
- **C.** Negative
- **D.** Infinite

Q3

If mass of a charged particle is doubled, cyclotron frequency becomes:

- **A.** Doubled
- **B. Halved ✓ Correct**
- **C.** Quadrupled
- **D.** One fourth

Q4

In uniform circular motion, average velocity is equal to same as:

- **A.** Linear velocity
- **B.** Average velocity
- **C. Uniform velocity ✓ Correct**
- **D.** Instantaneous velocity

Q5

Electric field intensity of infinite charge sheet:

- **A.** E = 2σ/ε■
- **B.** E = 1/ε■
- **C.** E = σ/ε■
- **D. σ/(2ε■) ✓ Correct**

Q6

Stability of radioactivity element is determined by:

- **A. Decay constant ✓ Correct**
- **B.** Rate of the radioactive decay
- **C.** Time of decay
- **D.** Number of radioactive nuclei

Q7

1 kWh is equal to:

- **A.** 3.6 MW
- **B. 3.6 MJ ✓ Correct**
- **C.** 3.6 kJ
- **D.** 3.6 GJ

Q8

Rectifier converts:

- **A. Alternating voltage into Direct voltage ✓ Correct**
- **B.** Direct voltage into Alternating voltage
- **C.** Alternating current into Direct current
- **D.** Direct current into Alternating current

Q9

Basic requirement for induced emf is:

- **A.** Flux linked to coil
- **B.** No flux linked to coil
- **C.** Coil should be circular
- **D. Change in flux links to the coil ✓ Correct**

Q10

Product of force and time is equal to:

- **A.** Change in displacement
- **B. Change in momentum ✓ Correct**
- **C.** Change in velocity
- **D.** Change in acceleration

Q11

Resistivity of copper is:

- **A. 1.54 × 10■■ Ω·m ✓ Correct**
- **B.** 1.34 × 10■■ Ω·m
- **C.** 1.45 × 10■■ Ω·m
- **D.** 1.64 × 10■■ Ω·m

Q12

Bursting of tire is an example of which process?

A. Adiabatic ✓ Correct

B. Isochoric

C. Isothermal

D. Isobaric

Q13

Direction of angular velocity can be found by:

A. Right-hand rule ✓ Correct

B. Fleming's left-hand rule

C. Lenz's law

D. Faraday's law

Q14

When star moves towards the earth, the shift is known as:

A. Blue shift ✓ Correct

B. Red shift

C. Compton shift

D. None

Q15

Angle at which maximum range of projectile is achieved?

A. 0°

B. 45° ✓ Correct

C. 90°

D. 76°

Q16

Which of the following spectrum is produced when atoms are in the gaseous state?

A. Continuous spectrum

B. Band spectrum

C. Line spectrum ✓ Correct

D. Absorption spectrum

Q17

Conductance is reciprocal of resistance, its unit is:

A. Siemen ✓ Correct

B. Ohm

C. Ampere

D. Watt

Q18

During rubbing of hands, internal energy:

- **A. Increases ✓ Correct**
- **B.** Decreases
- **C.** Zero
- **D.** Remains same

Q19

If velocity is doubled, what happens to kinetic energy:

- **A.** Remains same
- **B.** Doubles
- **C. Quadruples ✓ Correct**
- **D.** Halves

Q20

On a clear day at noon, intensity of solar energy is:

- **A. 1.4 kW/m^2 ✓ Correct**
- **B.** 1.2 kW/m^2
- **C.** 1 kW/m^2
- **D.** 1.3 kW/m^2

Q21

Work done is positive if angle is:

- **A. $0° < \theta < 90°$ ✓ Correct**
- **B.** $0° < \theta < 180°$
- **C.** $90° < \theta < 270°$
- **D.** None of these

Q22

The electric potential at a distance r is given by the formula:

- **A. $V = kq/r$ ✓ Correct**
- **B.** 1/4πε■qr
- **C.** Ed
- **D.** Ir

Q23

Which of the following act as a semiconductor?

- **A.** Iron
- **B. Silicon ✓ Correct**
- **C.** Zinc
- **D.** Tungsten

Q24

The fraction change in resistance per kelvin is:

A. Coefficient of conductivity

B. Temperature coefficient of resistance ✓ Correct

C. Coefficient of conductance

D. Coefficient of resistivity

Q25

Quantized momentum was explained by:

A. Einstein

B. Bohr ✓ Correct

C. de-Broglie

D. Rutherford

Q26

Angle between a and r is:

A. 90° ✓ Correct

B. 80°

C. 45°

D. 60°

Q27

EMF induced is zero when plane of coil is:

A. Perpendicular to B ✓ Correct

B. Parallel to B

C. Antiparallel

D. None of these

Q28

The spectra obtained by absorption or emission of radiation by atoms of the gas:

A. Atomic spectra ✓ Correct

B. Molecular spectra

C. Band spectra

D. Continuous spectra

Q29

Two waves identical traveling in the same medium are superposed:

A. Interference ✓ Correct

B. Diffraction

C. Reflection

D. Refraction

Q30

The work done by centripetal force is:

- **A.** Maximum
- **B.** Minimum
- **C. Zero ✓ Correct**
- **D.** Negative

Q31

Which of the following has highest specific heat capacity?

- **A.** Copper
- **B.** Iron
- **C. Water ✓ Correct**
- **D.** Glass

Q32

In photoelectric effect, stopping potential depends upon:

- **A.** Intensity of light
- **B. Frequency of light ✓ Correct**
- **C.** Speed of light
- **D.** Amplitude of light

Q33

Power of a lens is measured in:

- **A.** Meter
- **B. Dioptre ✓ Correct**
- **C.** Candela
- **D.** Watt

Q34

The SI unit of magnetic flux is:

- **A.** Tesla
- **B. Weber ✓ Correct**
- **C.** Henry
- **D.** Farad

Q35

Half-life of a radioactive element is 10 days. After 30 days, fraction remaining is:

- **A.** 1/2
- **B.** 1/4
- **C. 1/8 ✓ Correct**
- **D.** 1/16

Q36

Which type of wave cannot be polarized?

A. Radio waves

B. Light waves

C. Sound waves ✓ Correct

D. X-rays

Q37

The energy stored in a capacitor is:

A. QV

B. QV/2 ✓ Correct

C. Q^2V

D. 2QV

Q38

Which phenomenon proves the transverse nature of light?

A. Interference

B. Diffraction

C. Polarization ✓ Correct

D. Refraction

Q39

Absolute zero temperature is equal to:

A. 0°C

B. -173°C

C. -273°C ✓ Correct

D. -373°C

Q40

The process of converting AC into pulsating DC is called:

A. Amplification

B. Modulation

C. Rectification ✓ Correct

D. Filtration

Section 2: Chemistry

Q41

Most molecular crystals melt below:

- **A. 200°C ✓ Correct**
- **B.** 280°C
- **C.** 330°C
- **D.** 480°C

Q42

How much heavier is the H-atom as compared to electron?

- **A.** 1937
- **B.** 1737
- **C. 1837 ✓ Correct**
- **D.** 1637

Q43

The value of equilibrium constant for the decomposition of ozone at 25°C:

- **A. 10■■ ✓ Correct**
- **B.** 10■■
- **C.** 10^{2}■
- **D.** 10■■

Q44

A reaction is first order with respect to A and third order with respect to B, the rate equation is:

- **A.** Rate = $K[A][B]^2$
- **B.** Rate = $K[A]^2[B]$
- **C. Rate = $K[A][B]^3$ ✓ Correct**
- **D.** Rate = $K[A]$

Q45

For a reaction A $\rightarrow$ Product, if concentration of A is doubled, product increases 4 times. The order of reaction will be:

- **A.** Zero
- **B.** 1st
- **C. 2nd ✓ Correct**
- **D.** 3rd

Q46

Select the spontaneous reaction from the following:

- **A.** Reaction of nitrogen and oxygen
- **B. Emission of α-particles from polonium ✓ Correct**
- **C.** Thermal decomposition of marble
- **D.** Burning of paper

Q47

Select the exothermic reaction among the following:

- **A.** Thermal composition of P■O■
- **B.** Formation of atmospheric nitric oxide
- **C. Burning of methane ✓ Correct**
- **D.** Sublimation of dry ice

Q48

The oxidation number of Cl in HClO■ is:

- **A.** +1
- **B. +3 ✓ Correct**
- **C.** +5
- **D.** +7

Q49

Bond order is equal to:

- **A.** Difference between number of bonding electrons and antibonding electrons
- **B.** The sum of number of bonding electrons and antibonding electrons
- **C. Half of the difference between number of bonding electrons and antibonding electrons ✓ Correct**
- **D.** Quarter of the difference between number of bonding electrons and antibonding electrons

Q50

Which of the following molecule has non-polar covalent bond?

- **A.** Methyl chloride
- **B.** Methanol
- **C. Carbon dioxide ✓ Correct**
- **D.** Tetrachloromethane

Q51

Which of the following alcohols will give oily layer upon heating?

- **A.** 3-Methyl-2-butanol
- **B. 2-Methyl-2-butanol ✓ Correct**
- **C.** 2-Butanol
- **D.** Ethyl alcohol

Q52

Formalin is a mixture of:

- **A. 40% HCHO, 8% CH■OH, 52% H■O ✓ Correct**
- **B.** 40% CH■CHO, 8% CH■OH, 50% H■O
- **C.** 40% CH■CHO, 8% C■H■OH, 50% H■O
- **D.** 45% HCHO, 8% C■H■OH, 50% H■O

Q53

Polymerization of HCHO and CH■CHO occurs in the presence of:

- **A. Conc. H■SO■ ✓ Correct**
- **B.** Dil. H■SO■
- **C.** Conc. H■CO■
- **D.** Dil. H■CO■

Q54

The boiling point of carboxylic acids are relatively high due to:

- **A.** Increased molecular mass
- **B.** Intermolecular carbon bonding
- **C.** Intermolecular oxygen bonding
- **D. Intermolecular hydrogen bonding ✓ Correct**

Q55

Carboxylic acid reacts with metal carbonate. The colorless gas evolved is:

- **A.** H■
- **B.** O■
- **C. CO■ ✓ Correct**
- **D.** CO

Q56

Which of the following compound will react with 2,2-dimethyl propanoic acid to produce ester?

- **A.** Ethanoic acid
- **B.** Propane
- **C. Ethanol ✓ Correct**
- **D.** Propanone

Q57

The most abundant protein in animal kingdom forming 25–35% of body proteins is/are:

- **A.** Derived proteins
- **B. Simple proteins ✓ Correct**
- **C.** Compound proteins
- **D.** Conjugated proteins

Q58

Protein present in connective tissues of human body is:

- **A.** Phosphoprotein
- **B. Collagen ✓ Correct**
- **C.** Lecithin
- **D.** Cholesterol

Q59

In human body Ferritin is used to store:

- **A.** P
- **B.** Ca
- **C. Fe ✓ Correct**
- **D.** Zn

Q60

In human body Ceruloplasmin acts as a carrier of:

- **A.** O■
- **B.** Zn
- **C. Cu ✓ Correct**
- **D.** Fe

Q61

Predict the element with lowest ionization energy:

- **A. Mg ✓ Correct**
- **B.** Ar
- **C.** Si
- **D.** P

Q62

Select the group II element which has the highest atomic radius?

- **A.** Be
- **B.** Ca
- **C. Ba ✓ Correct**
- **D.** Sr

Q63

Which of the following element has least boiling point?

- **A.** Li
- **B.** Na
- **C.** K
- **D. Cs ✓ Correct**

Q64

2-pentanone and 3-pentanone are examples of:

- **A.** Metamers
- **B.** Functional group isomers
- **C.** Tautomerism
- **D. Position isomers ✓ Correct**

Q65

Which of the following will show optical isomerism?

- **A.** 2,2-Dihydroxy propanoic acid
- **B.** 2-Hydroxy butanedioic acid
- **C.** Butanoic acid
- **D. 2-Hydroxy propanoic acid ✓ Correct**

Q66

Which of the following is an electrophile in sulphonation?

- **A.** SO■H
- **B. SO■ ✓ Correct**
- **C.** SO■
- **D.** SO■■

Q67

Acetylene reacts with hydrogen bromide to form:

- **A.** 2,2-Dibromoethane
- **B. 1,1-Dibromoethane ✓ Correct**
- **C.** 1,2-Dibromoethane
- **D.** 1,1,2,2-Tetrabromoethane

Q68

Generally the decreasing order of reactivity is:

- **A.** Alkane > Alkynes > Alkene
- **B.** Alkene < Alkynes < Alkanes
- **C. Alkene > Alkynes > Alkanes ✓ Correct**
- **D.** Alkynes < Alkanes < Alkene

Q69

Racemic mixture formed in SN1 reaction is due to presence of empty:

- **A.** s-orbital
- **B. p-orbital ✓ Correct**
- **C.** sp^3-orbital
- **D.** sp^2-orbital

Q70

The order of reactivity of alcohol with respect to cleavage of C–O bond:

- **A. Tertiary > Secondary > Primary ✓ Correct**
- **B.** Primary > Tertiary > Secondary
- **C.** Secondary > Tertiary > Primary
- **D.** Tertiary < Secondary < Primary

Q71

NaOH + H■SO■ → Na■SO■ + H■O. When 40g of NaOH reacts with 49g of H■SO■, the number of water molecules produced are:

- **A. 6.02×10^{23} ✓ Correct**
- **B.** 6.02×10^{2}■
- **C.** 1.2044×10^{23}
- **D.** 1.204×10^{2}■

Q72

The number of electrons in a subshell can be calculated by the formula:

- **A.** $2l + 1$
- **B.** $2l + 2$
- **C.** $2(2 + l)$
- **D. $2(2l + 1)$ ✓ Correct**

Q73

When 5d orbital is complete, the entering electron goes into:

- **A.** 6s
- **B. 6p ✓ Correct**
- **C.** 6d
- **D.** 6f

Q74

Which of the following relationship is correct for kinetic energy of gases?

- **A.** K.E = 2RT/3NA
- **B. K.E = 3RT/2NA ✓ Correct**
- **C.** K.E = 2mNT/2R
- **D.** K.E = 2NA/3RT

Q75

The fourth state of matter is:

- **A.** Solid
- **B.** Liquid
- **C.** Gaseous liquid
- **D. Plasma ✓ Correct**

Q76

Which factor does not affect the rate of evaporation?

- **A. Amount of liquid ✓ Correct**
- **B.** Surface area
- **C.** Temperature
- **D.** Inter-molecular forces

Q77

Which of the following is responsible for the low boiling point of diethyl ether as compared to water?

- **A. Weak intermolecular forces ✓ Correct**
- **B.** High surface tension
- **C.** Low vapour pressure
- **D.** High vapour pressure

Q78

Hydrogen bonding does not exist in:

- **A.** Proteins
- **B.** Alcohols
- **C.** Liq HF
- **D. SiH■ ✓ Correct**

Q79

The factor that affects the lattice energy is:

A. Charge to size ratio ✓ Correct

B. Shielding effect

C. Crystal structure

D. Atomic radius

Q80

The shape of the crystal lattice depends upon:

A. Size of particles

B. Nature of particles

C. Arrangement of particles ✓ Correct

D. Size of unit cell

Section 3: English

Q81

Identify the sentence with correct punctuations:

A. Ali received a Parker pen, Hamza: a watch.

B. Ali received a Parker pen; Hamza a watch.

C. Ali received a Parker pen; Hamza: a watch. ✓ Correct

D. Ali received a Parker pen; Hamza, a watch.

Q82

Fill in the correct articles: ___ female lion is called ___ lioness. She has no mane.

A. A, the

B. The, a ✓ Correct

C. The, the

D. A, a

Q83

'Whatever you order for dinner' is fine with me. The underlined part is:

A. Noun clause ✓ Correct

B. Adjective clause

C. Adverb clause

D. Independent clause

Q84

After midnight, the ghost will come out of the haunted attic to scare the people. The sentence is:

A. Simple ✓ Correct

B. Compound

C. Complex

D. Compound-Complex

Q85

I suggest that you _____ visit a doctor.

A. Must

B. Should ✓ Correct

C. Ought to

D. Have to

Q86

Choose the correctly structured sentence.

A. Had he been there, he would see us.

B. Had he been there, he would had seen us.

C. Had he been there, he would have seen us. ✓ Correct

D. Had he had been there, he would have seen us.

Q87

Identify the sentence with correct structure.

A. We noticed the boy, walking down the street.

B. We noticed, the boy walk down the, street.

C. We noticed, that the boy was walking down the street.

D. We noticed the boy, who was walking down the street. ✓ Correct

Q88

He circled _____ whole world on his ship and on _____ island he encountered _____ fierce giant.

A. the, an, the

B. the, a, a

C. the, an, a ✓ Correct

D. the, an, an

Q89

Choose the sentence with correct order of adjectives.

A. I met, young two, beautiful, British girls at the airport.

B. I met two beautiful, young, British girls at the airport.

C. I met two, British, young, beautiful girls at the airport.

D. I met two young, beautiful British girls at the airport. ✓ Correct

Q90

Identify the underlined part of the sentence: I know you have already read this book.

- **A.** Independent clause
- **B.** Adjective clause
- **C.** Adverb clause
- **D. Noun clause ✓ Correct**

Q91

Choose the sentence with correct punctuations.

- **A.** "go then," Said the ant, "And dance winter away."
- **B.** "Go then" Said the Ant, and "dance winter away."
- **C. "Go then," said the ant, "and dance winter away." ✓ Correct**
- **D.** "go then". Said the Ant, "and dance Winter away."

Q92

Which of the following verbs takes the preposition 'from' with it?

- **A. Abstain ✓ Correct**
- **B.** Accused
- **C.** Desirous
- **D.** Ignorant

Q93

According to scientists, the sun is _____ and we can find the early discoveries made by scientists in black and white on _____. Choose correctly spelt homophones:

- **A.** Stationery, Stationary
- **B. Stationary, Stationery ✓ Correct**
- **C.** Stationary, Stationary
- **D.** Stationery, Stationary

Q94

Choose the correct preposition. One hour _____ glorious life is worth an age without a name.

- **A.** In
- **B. Of ✓ Correct**
- **C.** To
- **D.** With

Q95

Clive had been only a few months in the army when announced that peace had been concluded between great britain and france. The sentence contains errors of:

A. Comma

B. Comma and full stop

C. Comma and capitalization

D. Capitalization ✓ Correct

Section 4: Biology

Q96

Induced fit model of enzyme activity suggests that an enzyme:

A. Cannot modify its active sites

B. Can bind to a single substrate

C. Can catalyze related specific reaction ✓ Correct

D. Usually belongs to non-regulatory enzyme

Q97

Select the one which is used by hexokinase as an activator:

A. Cu■■

B. Zn■■

C. Mg■■ ✓ Correct

D. Fe■■

Q98

The linkage between substrate molecule and active site is:

A. Ionic bonding

B. Disulfide bonding

C. Hydrogen bonding ✓ Correct

D. Covalent bonding

Q99

Activated pepsin in which polypeptide fragment is removed is an example of:

A. Apo-enzyme

B. Holoenzyme

C. Regulatory enzyme

D. Non-regulatory enzyme ✓ Correct

Q100

Reversible inhibitors affect the enzyme catalyzed reaction by:

A. Forming covalent bonds with active site

B. Destroying the globular structure

C. Forming weak linkages with enzymes ✓ Correct

D. Activating the catalytic sites

Q101

What is the required medium for the maximum activity of the given enzymes? (Lipase / Pepsin)

A. Acidic / Basic

B. Acidic / Acidic

C. Basic / Acidic ✓ Correct

D. Basic / Basic

Q102

What distinguishes the concept of special creation from concept of evolution?

A. Life is not the product of sudden creative act

B. Unicellular prokaryotes might be the life ancestors

C. Rely on inspiration and meditation for life origin ✓ Correct

D. Life results from innumerable changes

Q103

Which of the following idea is a part of Lamarckism?

A. Use and disuse of organs ✓ Correct

B. Survival of the fittest

C. Origin of species

D. Variation

Q104

Choose the correct pair for homology?

A. Wings of birds and wings of butterfly

B. Forelimbs of birds and fins of whales ✓ Correct

C. Leaves of pines and cactus

D. Gills of fish and lungs of humans

Q105

Which chronological sequence is correct among the classes of vertebrates as evidence of evolution?

- **A.** Birds → fish → amphibians → reptiles
- **B.** Reptiles → birds → fishes → amphibians
- **C. Fish → amphibians → reptiles → birds ✓ Correct**
- **D.** Amphibians → reptiles → birds → fish

Q106

The reduction of population carrying a specific allele and genotype due to natural disaster is called:

- **A.** Mutation
- **B. Bottle neck effect ✓ Correct**
- **C.** Founder effect
- **D.** Speciation

Q107

Which of the following increases variation within a gene pool?

- **A.** Adaptation
- **B.** Genetic drift
- **C. Gene mutation ✓ Correct**
- **D.** Neutral selection

Q108

Identify the location of mitral valve in human heart:

- **A.** Between right ventricle and pulmonary trunk
- **B.** Between left ventricle and aorta
- **C.** Between right atrium and right ventricle
- **D. Between left atrium and left ventricle ✓ Correct**

Q109

Which of the following components are present both in blood and lymph?

- **A.** O■, water, large proteins
- **B.** RBCs, electrolytes, amino acids
- **C.** CO■, RBCs and water
- **D. CO■, glucose and water ✓ Correct**

Q110

Blood flow in vessels least depends on:

A. Total cross sectional area of vessels

B. Blood pressure

C. Skeletal muscle contraction ✓ Correct

D. Heart beat

Q111

Which T-lymphocytes do not directly attack invading microbes?

A. Natural killer cell

B. Macrophages

C. B-lymphocytes ✓ Correct

D. Neutrophils

Q112

A woman cuts her finger while working in kitchen. Which chemical substance will cause inflammation around wound?

A. Interleukin – I

B. Histamine ✓ Correct

C. Interleukin – II

D. Heparin

Q113

Choose the mismatched:

A. Acetylcholine – gastrin

B. Enterokinase – lipase

C. Emulsification – fat droplets

D. Chylomicrons – cholesterol + carbohydrates ✓ Correct

Q114

Which protein is present in the walls of alveoli?

A. Collagen ✓ Correct

B. Hemoglobin

C. Myoglobin

D. Myosin

Q115

Which of the following acts as an enzyme activator in human digestive system?

- **A.** Erepsin
- **B. Enterokinase ✓ Correct**
- **C.** Pepsinogen
- **D.** Trypsinogen

Q116

How would you classify a bacterium having a group of 2 or more flagella at one pole?

- **A. Lophotrichous ✓ Correct**
- **B.** Amphitrichous
- **C.** Monotrichous
- **D.** Peritrichous

Q117

Which of the following bacteria is used for developing vaccine against tuberculosis?

- **A.** Mycobacterium tuberculosis
- **B.** E. coli
- **C.** Salmonella typhi
- **D. Mycobacterium bovis ✓ Correct**

Q118

If a bacteria can grow in the presence or absence of oxygen it would be a(n):

- **A.** Aerobic
- **B.** Anaerobic
- **C. Facultative anaerobe ✓ Correct**
- **D.** Microaerophilic

Q119

Which testicular cells in a human male with high testosterone level will start the endocrine activity?

- **A.** Interstitial cell
- **B.** Sertoli cells
- **C.** Spermatogonial cells
- **D. Leydig cells ✓ Correct**

Q120

Which layer of uterus is composed of contractile muscles?

- **A.** Endometrium
- **B.** Perimetrium
- **C. Myometrium ✓ Correct**
- **D.** Both endo and myometrium

Q121

One of the endocrine glands responsible for failure in ovulation is:

- **A. Pituitary ✓ Correct**
- **B.** Placenta
- **C.** Thyroid
- **D.** Adrenal

Q122

Identify the exact function of ATP hydrolysis during muscle contraction:

- **A.** Formation of cross bridges
- **B. Breaking of cross bridges ✓ Correct**
- **C.** Displacement of tropomyosin
- **D.** Release of Ca^{2}■ ions from sarcoplasmic reticulum

Q123

The reason behind the slow healing of cartilage is:

- **A.** Absence of minerals in matrix
- **B.** Absence of nervous supply
- **C. Absence of blood supply ✓ Correct**
- **D.** Loose arrangement of cells

Q124

H zone of sarcomere consists of:

- **A.** Proteins supporting myosin filaments
- **B.** Myosin overlapped by actin
- **C.** Only actin filament
- **D. Only myosin filament ✓ Correct**

Q125

Which cell is most important to cause bone resorption?

- **A.** Osteoblast
- **B. Osteoclast ✓ Correct**
- **C.** Osteocyte
- **D.** Fibroblast

Q126

Activity of which organ would be compromised with impaired function of smooth muscles?

- **A.** Heart
- **B.** Lungs
- **C. Stomach ✓ Correct**
- **D.** Tongue

Q127

The phenomena of gene linkage rejects the concept of:

A. Law of segregation

B. Polygenic traits

C. Epistasis

D. Law of independent assortment ✓ Correct

Q128

Which of the following is used to determine linkage between two gene pairs?

A. Test cross ✓ Correct

B. Back cross

C. Monohybrid cross

D. Dihybrid cross

Q129

The process by which unwanted structures are engulfed and digested within the lysosome is termed as:

A. Autophagy ✓ Correct

B. Autolysis

C. Endocytosis

D. Exocytosis

Q130

Fluidity of biological membranes is maintained by:

A. Glycoprotein

B. Saturated fatty acid

C. Fatty acids

D. Phospholipids ✓ Correct

Q131

Which of the following independently reproduces in the cell?

A. Golgi complex

B. Nucleus

C. Chloroplast ✓ Correct

D. Endoplasmic reticulum

Q132

Which statement best describes difference of prokaryotic cell from eukaryotic cell?

A. Prokaryote: Membrane bound organelle is present / Eukaryote: Membrane bound organelle is absent

B. Prokaryote: Ribosomal subunits are 50S & 30S / Eukaryote: Ribosomal subunits are 60S & 40S ✓ Correct

C. Prokaryote: DNA is double stranded / Eukaryote: DNA is circular

D. Prokaryote: Multicellular / Eukaryote: Multicellular and unicellular

Q133

Skeletal muscle cells are having:

A. Less SER & more RER

B. More SER & less RER ✓ Correct

C. Equal amount of SER & RER

D. No SER & RER

Q134

Which one is the correct direction of action potential along the neuron?

A. Axon → post synaptic terminal ✓ Correct

B. Schwann cells → pre synaptic terminal

C. Cell body → pre synaptic terminal

D. Dendrite → post synaptic terminal

Q135

Concentration of Na■ and K■ ions during resting membrane potential is: (Inside Neuron / Outside Neuron)

A. K■ 30 times / Na■ 10 times

B. K■ 10 times / Na■ 30 times ✓ Correct

C. Na■ 10 times / K■ 30 times

D. Na■ 30 times / K■ 10 times

Q136

A brief phase in which neuron regains its original ionic distribution and polarity is called:

A. Repolarization

B. Synaptic delay

C. Saltatory period

D. Refractory period ✓ Correct

Q137

Parts of brain that comprise the limbic system:

- **A.** Thalamus, amygdala, hippocampus
- **B. Hypothalamus, amygdala, hippocampus ✓ Correct**
- **C.** Hypothalamus, corpus callosum, hippocampus
- **D.** Thalamus, hypothalamus, pons

Q138

Ions involved in synaptic transmission of nerve impulse:

- **A.** K■, Ca^{2}■, Na■
- **B.** Na■, K■
- **C. Na■, Ca^{2}■ ✓ Correct**
- **D.** K■, Ca^{2}■

Q139

Correct sequence of the four elements of a neural pathway in simple-reflex circuit:

- **A. Sensory neuron → associative neuron → motor neuron → muscles ✓ Correct**
- **B.** Muscles → sensory neuron → associative neuron → motor neuron
- **C.** Motor neurons → associate neuron → sensory neuron → muscles
- **D.** Associate neuron → sensory neuron → motor neuron → muscles

Q140

In humans memory storage for behavior and coordination is well developed in the:

- **A.** Amygdala
- **B.** Cerebrum
- **C.** Medulla
- **D. Cerebellum ✓ Correct**

Q141

Choose the correct pairing: (Chemical Nature / Hormones / Glands)

- **A. Proteins / Insulin / Pancreas ✓ Correct**
- **B.** Steroid / Cortisone / Thymus
- **C.** Catecholamine / Estrogen / Adrenal medulla
- **D.** Amino acid derivative / Thyroxine / Parathyroid

Q142

Which phyla exhibit first true coelom?

- **A.** Cnidaria
- **B. Annelida ✓ Correct**
- **C.** Mollusca
- **D.** Platyhelminthes

Q143

Which invertebrate has exceptionally large and complex brain with highly developed capabilities to learn and remember?

- **A.** Snail
- **B.** Cattle fish
- **C. Octopus ✓ Correct**
- **D.** Giant squid

Q144

Arthropods constitute 54% of kingdom Animalia. What is the reason for their success?

- **A. Diversity and articulation of exoskeleton ✓ Correct**
- **B.** Existence of haemocoel
- **C.** Well-developed reproductive system
- **D.** Bilateral symmetry

Q145

Which of the following virus is having double stranded RNA in its core?

- **A.** Small pox
- **B.** Mild rash virus
- **C. Diarrhea virus ✓ Correct**
- **D.** Rubella virus

Q146

Pore size of porcelain filter ranges between:

- **A.** 10 – 100 nm
- **B. 100 – 1000 nm ✓ Correct**
- **C.** 10 – 10 µm
- **D.** 100 – 1000 µm

Q147

Reverse transcriptase converts:

- **A.** ds RNA to ss RNA
- **B.** ss RNA to ds RNA
- **C. ss RNA to ds DNA ✓ Correct**
- **D.** ss DNA to ds DNA

Q148

Which symptoms disappear after few months in an asymptomatic carrier of HIV?

- **A.** Loss of memory
- **B. Swollen lymph gland ✓ Correct**
- **C.** Weight loss
- **D.** Night sweats

Q149

The primary purpose of photosystem-II in thylakoid membrane is:

- **A.** Synthesis of ATP
- **B. Oxidation of water ✓ Correct**
- **C.** Reduction of NADP+ to NADPH
- **D.** Oxidation of cytochrome

Q150

In glycolysis glucose is:

- **A. Oxidized into pyruvic acid ✓ Correct**
- **B.** Reduced into pyruvate
- **C.** Broken down to G3P
- **D.** Reduced to NADPH